



✓ **SECTION A ($2 \times 5 = 10$ Marks)**

(Each answer carries 2 marks)

Q1. [CO3] What is a WordPress theme?

Answer:

A WordPress theme is a collection of template files, stylesheets, and assets that determine the design, layout, and appearance of a WordPress website. Themes control how content is displayed without changing the core functionality of WordPress. Users can switch themes anytime without affecting website data.

Q2. [CO3] How can we customize a WordPress theme with CSS?

Answer:

A WordPress theme can be customized with CSS using:

1. **Customizer** → **Additional CSS** option to write custom styles.
2. Editing the **style.css** file inside the theme folder.
3. Using **child theme** CSS to add changes without modifying the main theme.
4. Installing custom CSS plugins like “Simple Custom CSS”.

Q3. [CO3] What is functions.php file in a theme?

Answer:

functions.php is a theme-specific PHP file that allows developers to add custom features and extend WordPress functionality. It works like a plugin and can include code for menus, widgets, shortcodes, theme support, scripts, and hooks.

Q4. [CO3] What are custom taxonomies in WordPress?

Answer:

Custom taxonomies are user-defined classification systems used to group and organize content beyond default categories and tags. Example: A movie website may use taxonomies like *Genre*, *Actor*, *Language*, etc. They can be **hierarchical** (like categories) or **non-hierarchical** (like tags).

Q5. [CO3] What is custom posts in WordPress?

Answer:

Custom Post Types (CPT) are content types created in addition to default “Posts” and “Pages.” They allow storing specialized content such as *Portfolio*, *Events*, *Products*, *Testimonials*, *Jobs*, etc. They are created using `register_post_type()` in functions.php or via plugins like CPT UI.

✓ **SECTION B ($3 \times 7 = 21$ Marks)**

(Each answer carries 7 marks)

Q1. [CO3] Explain how we can setup Menus in WordPress?

Answer:

WordPress provides a built-in Menu Management System to create and display navigation menus.

Steps to Set Up a Menu:

1. Go to **Appearance** → **Menus** from dashboard.
2. Create a new menu and give it a name (e.g., Header Menu).
3. Add items such as Pages, Posts, Categories, Custom Links, etc.
4. Arrange using drag-and-drop to create hierarchy (submenus).
5. Assign menu to locations like Header, Footer, Sidebar based on theme.

6. Save menu and preview on frontend.

Advanced Menu Options:

- ✓ Mega menus using plugins
- ✓ Adding icons with CSS or FontAwesome
- ✓ Conditional menus (logged-in/out users)
- ✓ Menu customization via WordPress Customizer

Menus improve website navigation, SEO, and user experience.

Q2. [CO3] Discuss anatomy of WordPress theme development.

Answer:

A WordPress theme has a defined structure containing mandatory and optional files.

Main Components of a Theme:

File	Purpose
style.css	Main stylesheet + theme metadata
index.php	Fallback template for all pages
functions.php	Add custom features & hooks
header.php	Website and navigation
footer.php	Footer layout and scripts
single.php	Display single blog post
page.php	Display static page
archive.php	Display category/tag/date listings
sidebar.php	Sidebar widget area
comments.php	Comments layout
screenshot.png	Theme preview in dashboard

Theme Development Steps:

1. Create folder in /wp-content/themes/
2. Add mandatory files: style.css, index.php
3. Add template hierarchy files (single.php, page.php, etc.)
4. Register menus, widgets in functions.php
5. Use WordPress hooks, filters, template tags
6. Make theme responsive, SEO-friendly, and translation-ready

Q3. [CO3] Discuss the use of the Loop in WordPress.

Answer:

The WordPress Loop is a PHP code structure used to display posts fetched from the database. It checks if posts exist and outputs them dynamically.

Basic Syntax:

```
<?php if ( have_posts() ) :  
    while ( have_posts() ) : the_post(); ?>  
        <h2><?php the_title(); ?></h2>  
        <?php the_content(); ?>  
<?php endwhile; endif; ?>
```

Purpose of the Loop:

- ✓ Display posts on blog page, archive page, category page, etc.
- ✓ Can be customized to show excerpts, featured images, metadata.
- ✓ Works with custom post types & custom queries (WP_Query).
- ✓ Allows conditional formatting (sticky posts, featured posts).

Example Uses:

- Displaying recent posts on homepage

- Filtering posts by category
- Showing related articles

The Loop is the core engine that powers dynamic content display in WordPress.

✓ SECTION C (3 × 11 = 33 Marks)

(Each answer carries 11 marks)

Q1. [CO3] Explain the components of a custom theme in WordPress.

Answer:

A **custom WordPress theme** is built from scratch or modified from a starter theme and contains template files, assets, and configuration code.

Key Components:

Component	Description
style.css	Required; contains theme name, author, version, styling
index.php	Main template file for displaying content
functions.php	Registers menus, widgets, scripts, CPT, taxonomies
header.php	Site header, logo, navigation, meta tags
footer.php	Closing HTML, footer widgets, scripts
sidebar.php	Sidebar widget area
template-parts/*	Reusable components such as post loops
screenshot.png	Preview image shown in dashboard
assets folder	CSS, JS, fonts, images
theme.json	(Block theme) controls global styling

Advanced Components:

- ✓ Custom blocks (Gutenberg)
- ✓ Customizer settings (theme options panel)
- ✓ Translation files (.po, .mo)
- ✓ Page templates (template-contact.php)
- ✓ Custom post types & taxonomies
- ✓ WooCommerce templates

A custom theme allows full control over design, responsive layout, and performance.

Q2. [CO3] Discuss third-party integration (API) in WordPress.

Answer:

Third-party API integration allows WordPress to fetch, send, or process external data such as payment, maps, SMS, social media, AI, etc.

Ways to Integrate API in WordPress:

1. **WP_HTTP / wp_remote_get()** to call REST APIs
2. JavaScript **fetch()** with **wp_localize_script**
3. **Custom plugin** for API handling
4. **Webhook integration** (for automation)
5. Using plugins for specific services (Stripe, MailChimp, Google Maps)

Common API Integrations:

- **Payment Gateways:** Stripe, Razorpay, PayPal
- **Social Media Login:** Google, Facebook, LinkedIn
- **AI/Chatbots:** OpenAI, Dialogflow
- **Email Services:** SendGrid, Mailchimp
- **Maps & Geolocation:** Google Maps API
- **SMS/OTP:** Twilio, Fast2SMS

- **Headless WordPress:** Using WP REST API with React, Next.js

Benefits:

- ✓ Enhances functionality without heavy coding
- ✓ Enables automation & real-time data
- ✓ Improves UX (e.g., autofill, live search, authentication)

Example Code Snippet:

```
$response = wp_remote_get('https://api.example.com/data');  
$data = json_decode(wp_remote_retrieve_body($response), true);
```

Q3. [CO3] Describe one installation and multiple blogs in WordPress.

Answer:

WordPress supports **Multisite Network**, a feature that allows running multiple websites (blogs) using a single WordPress installation.

Features of Multisite:

Feature	Details
Single WP Core	One installation controls all sites
Multiple Sites	Each site has own URL, themes, users
Shared Database	Same DB but separate tables per site
Super Admin Role	Manages entire network
Domain Options	Subdomains (site1.example.com) or subdirectories (example.com/site1)

Steps to Enable Multisite:

1. Add in wp-config.php:
define('WP_ALLOW_MULTISITE', true);
2. Install network setup from Tools → Network Setup
3. Configure subdomain or subdirectory mode
4. Update .htaccess and wp-config.php with generated rules
5. Access **Network Admin Dashboard** to create new sites

Use Cases:

- ✓ University website with departments
- ✓ News portal with multiple language editions
- ✓ SaaS multi-tenant platform
- ✓ Franchise websites under one system

Advantages:

- Centralized management
- Saves hosting resources
- Shared plugins and themes
- Single update for all sites

Disadvantages:

- Requires more server resources
 - If main site crashes, all sites go down
 - Plugin conflicts affect entire network
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